

Introduction

Dear reviewers,

Thank you for taking the time to review our manuscript and give valuable feedback on improvements. With your help, this article is greatly improved. Please see the corresponding sections below for a detailed response to each of your points. Overall, we have improved Figures 2 through 4 for readability. Additionally, the Discussion section has been enlarged and divided into subsections focused on: effects of resting length, nondimensionalization, comparison with other published BPA models (including the addition of Figure 5 and Table 3), implications for real-time control and simulation, and a discussion about the analogy of BPA artificial muscle compared to actual biological muscle. Reviewer comments are listed item-by-item, below, and our responses are on the proceeding indented lines.

Reviewer 1

1. Illustration of different lines in Fig. 2A and Fig. 2B should be added to these two subfigures, respectively. Double check such problems across this manuscript.

A legend has been added to Figure 2 and the text size has been enlarged on Figures 2-4.

2. The size of word in Figure 4 is too small, which should be revised.

The text size of Figures 2 through 4 has been increased.

3. There should be a space between the value and the unit, such as, 490 N. Check and revise across this manuscript.

We've used package siunitx with the “/,” space. We checked the document and corrected a few that did not. We can change the default options to a larger space, if needed, but would need approval from MDPI copy-editors.

4. A section of Conclusion should be added after the Discussion.

The last paragraph has now been organized into a Conclusion section.

5. The proposed models are largely empirical. How to demonstrate the versatility of the empirical model for Festo BPAs with different diameters?

This work applies to the 10 and 20 mm diameter muscles, which are the most common. We anticipate that these equations can be used with different coefficients for the other sizes (5mm and 40 mm), but data would need to be collected, and we are not set up to collect that data. We believe that because this provides detailed information on the most common diameter muscles used in the field of robotics, this paper contributes meaningfully to scientific knowledge without the inclusion of data and a fit for the 5 mm and 40 mm muscles and that such findings could be left to future research and publications.

6. What is the stiffness of the Festo BPA when a zero pressure is applied? Characterization of such a passive stiffness is demanded.

We agree that the stiffness characterization at zero pressure could be interesting, especially when considering a comparison of passive elements in biological muscles. However, we do not currently have equipment to perform this test, as it would require a unique tensile test system. The work that we have presented provides detailed information for active control of the artificial muscles, while the stiffness at zero pressure could only be found if the actuator is acted on by an outside force. We believe that since our work focuses on active control situations, it contributes meaningfully to scientific knowledge without the inclusion of passive stretch data at zero pressure

7. Experiments are limited to Festo BPAs of two diameters; effects of manufacturing variations are not analyzed. Add necessary discussion on these influences.

We have indeed seen some variation in data that we attribute to manufacturing variations, this is mentioned some in the discussion but was not expanded upon. We have added to the discussion paragraph about this subject and it now reads as follows:

Additionally, manufacturing variations seem to play an important factor in the behavior of these actuators. It remains to be seen how ϵ_{620} can be known a priori. We still suspect that it is a function of l_{rest} , but it might also be a function of other factors such as product batch number or UV light exposure [19](p. 15); [21](p. 6); [22, p. 13](p. 13). We found there

Commented [AH1]: Need to revert 5, 6, and 8 back. No need to explain the reason for not collecting specific data within the paper.

to be larger variations in the 10 mm BPAs than in the 20 mm BPAs. Modeling work of ideal McKibben actuators describe how the relationship between contraction and initial length is a function of the total number of twists and the angle of twist in the usable muscle area [23]. It is unclear if there are variations in the angle and number of twists in the Festo BPAs due to manufacturing. Uncovering the relationship between ϵ_{620} and l_{rest} for the Festo DMSP/MAS artificial muscles, if it exists in a meaningful way, will require additional controlled tests. Similarly, it is not clear how the maximum force might change with manufacturing variations. This was reasonably corrected for by normalizing the equations to each individual's maximum contraction, but more nuanced details might emerge with more data collection. Not knowing these relationships a priori means there is still a need for robots and other systems to go through an iterative design stage if it is discovered that the system does not produce the torque or have the RoM that the design team expects. However, the data and models provided here should enable for faster redesigns with fewer additional measurements.

8. No analysis of force degradation over cyclic loading or long-term drift is provided. Add these data.

We agree that this would be interesting to look at and an important consideration for long-term use of these artificial muscles. We don't currently have a test setup for cycling the muscles under load and believe it is outside the scope of this paper, where data collection of this level would constitute a whole publication in itself. We have added the following to the discussion section of the paper.

The presented work is capable of predicting force output for 10 mm and 20 mm muscles with different initial muscle lengths, internal pressure, and contraction amount with minimal initial data collection. However, it remains to be seen if these relationships are maintained over the full lifecycle of the muscles. There are not yet any peer reviewed studies investigating the long-term behavior of these artificial muscles, though the Festo datasheet provides some indication of what can be expected [19](p.19). They report that the 40 mm BPA can complete 1 million cycles

at 6000 N (maximum rated load), or 10 million cycles at 4000 N. The service life of the other size BPAs is not reported, however it is mentioned that thermal stressing may reduce the lifespan and they should be inspected every 500,000 cycles for signs of aging [21](pp. 10–11). Future research should look at the behavior of these muscles over time.

References

19. Festo AG & Co. KG. Fluidic Muscle DMSP, 2026. [Product Datasheet]. Available online: https://www.festo.com/media/catalog/202851_documentation.pdf (accessed on 26 March 2026).
21. Festo AG & Co. KG. Fluidic Muscle DMSP/MAS, 2018. [Product Datasheet]. Available online: https://www.festo.com/net/en_us/SupportPortal/Files/732773/DMSP_MAS_2018-01f_8081496g1.pdf (accessed on 26 March 2026).

9. Quantitative comparison with existing models on the same dataset should be conducted to clarify the superiority of proposed model.

The reviewer is correct, we should compare to more than just the Festo tool, and have added comparisons to Sarosi [24], Sarosi and Fabulya [25], and Martens and Bolan [18]. We have added Table 3 to show a direct comparison of our model with these models, as well as a subsection in the Discussion that reads as follows:

5.3. Comparison with other published BPA Models

The proposed model was compared with the formulations Sárosi [24] , Sárosi and Fabulya [25] , and Martens and Boblan [18], to evaluate their predictive performance across actuator sizes and operating conditions (see Table 3 and Figure 5). For the $\phi 20$ mm BPA, parameters from [24] were used, corresponding to actuators with resting lengths of 200 mm and 400 mm. These were compared against the experimental data at $l_{rest} = 300$ mm and $l_{rest} = 450$ mm, respectively, because that is what gave the best fit. For the $\phi 10$ mm BPA with resting lengths of 257 mm and 233 mm , parameters from [25] were applied. In each case, the maximum contraction parameter (k_{max}) was determined numerically by solving $F(6.2 \text{ bar}, k_{max}) = 0$, ensuring consistency with the experimental maximum-pressure condition.

While the Sárosi [24] , Sárosi and Fabulya [25] models capture general trends in force–contraction behavior, discrepancies were

observed when applying parameters derived from different actuator lengths. In particular, the model tends to overpredict force at higher pressures and low amounts of contraction, suggesting sensitivity to geometric scaling effects not explicitly accounted for in the formulation.

The Martens and Boblan [18] model is derived from a geometric and material-based framework and provides a more physically grounded representation of actuator behavior. However, its accuracy depends strongly on precise geometric parameters, including braid angle, wall thickness, and fiber length. Small deviations in these parameters can lead to significant differences in predicted force, particularly at higher pressures.

The Martens and Boblan model was first evaluated at the same pressure–contraction points as in their experimental data [18](Table A1 and Table A2), indicating that it reproduces the general published force–map shape but is sensitive to the assumed initial geometry. Examining their $\phi 20$ mm BPA, for example, and using the explicitly stated $l_{rest} = 300$ mm and initial diameter $D_0 = 20$ mm significantly under-predicted the results compared to using the implied $l_{rest} = 296$ mm ([18](Table A1)) and $D_0 + H_0 = 21.8$ mm, where $H_0 = 1.8$ mm is the material thickness. The diameter adjustment can be implied from [23]. This adjustment for the diameter was also included for the $\phi 10$ mm BPA to give a better fit. As can be seen in Figure 5 and Table 3, the highest accuracy is observed when $\phi 20$ mm and $l_{rest} = 300$ mm, which is at or very near their experimental measurements. For the $\phi 10$ mm BPAs the accuracy decreases even though their experimentally measured $l_{rest} = 250$ mm is close to our $l_{rest} = 257$ mm and $l_{rest} = 233$ mm. The largest error occurs at $\phi 20$ mm and $l_{rest} = 450$ mm and the model starts to predict negative force. We have calculated that the $FPE \cdot dD/DL$ term becomes very large compared to the others [18](p. 5). This suggests that the model is sensitive to geometric parameter selection and interpretation, particularly with respect to initial length and diameter definitions. In comparison, the present normalized model demonstrates improved consistency across actuator lengths by incorporating experimentally derived scaling through F_{620} and ϵ_{620} . This normalization reduces sensitivity to geometric variation and enables a unified representation of actuator behavior, while maintaining predictive accuracy over the tested range of pressures and contractions.

18. Martens, M.; Boblan, I. Modeling the Static Force of a Festo Pneumatic Muscle Actuator: A New Approach and a Comparison to Existing Models. *Actuators* 2017, 6, 33.

<https://doi.org/10.3390/act6040033>.

24. Sárosi, J. New approximation algorithm for the force of fluidic muscles. In Proceedings of the Applied Computational Intelligence and Informatics (SACI), 2012 7th IEEE International Symposium on. IEEE, 2012, pp. 229–233.

25. Sárosi, J.; Fabulya, Z. Mathematical analysis of the function approximation for the force generated by pneumatic artificial muscle. *Scientific Bulletin of the Politehnica University of Timisoara, Transactions on Mechanics* 2012, 57, 59–64.

10. Why 620 kPa is chosen as the normalization reference? Whether other pressures can also be employed?

This is an excellent question. We attempted to answer this within the text, but perhaps it was not clear. This pressure was used because it is a common building supply pressure here (90 psi). Other pressures could relatively easily be used, like 600 kPa maximum rated pressure for the 20 mm BPA, or 800 kPa which is the maximum rated pressure for the 10 mm BPA. We have modified a paragraph in the nondimensionalization section of the discussion to read as follows:

In our work, $P_{620} = 620$ kPa is the maximum supply pressure of our building and for our system, and we note that other users of this actuator may use a different supply pressure. This equation will work for any pressure supplies below 620 kPa. Additionally, this supply pressure is not required for these equations, and this value was chosen for normalizing as a method that improves optimization by reducing the sensitivity of the term associated with pressure. This value is divided into the whole system and can easily be multiplied back in to eliminate the normalization if desired. We also anticipate the equations will also work for pressures up to 800 kPa (the maximum rated pressure for Festo $\phi 10$ mm BPAs), however, we do not have the data to confirm this.

11. Can the proposed model be embedded into real-time control or simulation? Clarify it in Discussion.

This model can indeed be embedded into real-time control and simulation. We have added a subsection to the Discussion to elaborate on this as follows:

5.4. Implications for Real-Time Control and Simulation

Although pneumatic muscle actuators exhibit hysteresis, thermodynamic effects, and other dynamic behavior, accurate and high-performance controller designs can still be achieved using a static force-map approach when the underlying static force model is accurate. Martens and Boblan explicitly note that BPA dynamics are often dominated by the air dynamics inside the actuator and the mass flow behavior of the valve, such that precise modeling of the static force characteristic remains crucial for torque control accuracy [18]. This view is consistent with more recent BPA control work, including a sensor-less torque-control interface for BPA-driven joints [26], stable force/position/stiffness control of antagonistic pneumatic-muscle joints [27,28], and sensor-less angle stiffness control of antagonistic BPA systems [29].

In this context, the present model is well suited for real-time use because it reduces the static isometric force prediction to a low-order normalized surface, $F^*(\epsilon^*, P^*)$, scaled by the resting-length-dependent term $F_{620}(l_{rest})$. This structure is computationally inexpensive to evaluate online and is also suitable for lookup-table implementation. While the present work does not model the full actuator-valve dynamics, it provides the static force relationship needed as a feedforward or inner-model component in real-time torque control, simulation, and preliminary design workflows. It is anticipated that improved real-time control for high speed and high precision systems will benefit more from improved modeling of airflow dynamics than actuator dynamics such as hysteresis, damping, and other behaviors.

12. Double check the format of references [19] and [20].

Thank you for pointing this out. The references have been corrected. Two additional Festo DMSP datasheet versions have also been referenced. These references are now:

19. Festo AG & Co. KG. Fluidic Muscle DMSP, 2026. [Product Datasheet]. Available online: https://www.festo.com/media/catalog/202851_documentation.pdf (accessed on 26 March 2026).

20. Lang, R. MuscleSIM, 2005. [Software]; not publicly available, Dept. TK-ES, Festo AG & Co. KG.

21. Festo AG & Co. KG. Fluidic Muscle DMSP/MAS, 2018. [Product Datasheet]. Available online: https://www.festo.com/net/en_us/SupportPortal/Files/732773/DMSP_MAS_2018--1f_8081496g1.pdf (accessed on 26 March 2026).

22. Festo AG & Co. KG. Fluidic Muscle DMSP/MAS, 2013. [Product Datasheet]. Available online: https://ftp.festo.com/Public/PNEUMATIC/SOFTWARE_SERVICE/Documentation/2015/EN/DMSP-MAS_EN.PDF (accessed on 26 March 2026).

Reviewer 2

A simple analysis of a product's performance and the difference between the manufacturer's specifications and the individual device. As a route to characterizing the actuator for use in an eventual product it is acceptable.

1. However, it would be informative for the authors to speculate the reasons for the deviations between their model and the manufacturer's. They identify *where* but not *why*. As experienced users of the devices they might be able to give the reader the benefit of their greater insight.

This is a great suggestion. We have added the following to the Effects of Resting Length subsection of the Discussion section:

The main cause for deviations from the Festo MuscleSIM tool is not fully clear. Festo AG & Co. KG [19] (p. 12) states that force will be reduced by approximately 10% for a minimum nominal length. In Festo AG & Co. KG [21] (p. 14) they say the force can deviate by 10% based on factors such as: production variation, material variation, and deviation from nominal length (l_{rest}). Festo AG & Co. KG [22] (p. 13) also notes that testing was done with BPAs where the length is ten times the uninflated diameter, and that theoretical force can increase by up to 10%. In each datasheet, the characteristic curves appear to be the same. As demonstrated above, the 100 mm ϕ 10 mm BPA should have much less force than what is shown in the curve. This has consistently been our experience and what motivated this research. It is unclear why the MuscleSIM tool then predicts higher force as resting length decreases. It could be a reversed sign coefficient in their internal curve-fit model. We have shown here that as the nominal length goes to zero, so does force. The reason for BPA

models predicting higher force than what is measured experimentally is a recurrent theme across many models as they do not account for the end effects associated with clamping the ends of pressure vessel. Nominally, the clamped ends fix the inner and outer diameter, which force steep gradients in stress and strain near the ends. Axial, radial, and circumferential strains deviate from analytic solutions to a cylindrical model at these boundaries. For constant pressures, the end effect region has the same length. Therefore, in shorter BPAs, the end effects dominate, greatly affecting the force-pressure-contraction curves, and the active force differs substantially from that of an assumed perfect cylinder.

References

19. Festo AG & Co. KG. Fluidic Muscle DMSP, 2026. [Product Datasheet]. Available online: https://www.festo.com/media/catalog/202851_documentation.pdf (accessed on 26 March 2026).
21. Festo AG & Co. KG. Fluidic Muscle DMSP/MAS, 2018. [Product Datasheet]. Available online: https://www.festo.com/net/en_us/SupportPortal/Files/732773/DMSP_MAS_2018-01f_8081496g1.pdf (accessed on 26 March 2026).
22. Festo AG & Co. KG. Fluidic Muscle DMSP/MAS, 2013. [Product Datasheet]. Available online: https://ftp.festo.com/Public/PNEUMATIC/SOFTWARE_SERVICE/Documentation/2015/EN/DMSP-MAS_EN.PDF (accessed on 26 March 2026).

2. One concern is: "*The improved Hill-muscle model of Thelen [22] shows an exponential term for a muscle's passive force-length property concentric loading.*" This model is specifically for aging muscles. In general the impact of the connective tissue on the curve is a step function. Do the authors have a more general reference for this?

The reviewer makes an excellent point. We have added a subsection titled Biomimetic Artificial Muscle Analogy to the Discussion and instead of Thelen we cite older work from Winters[30] and Chou and Hannaford[23]. The subsection now reads:

5.5. Biomimetic Artificial Muscle Analogy

BPA artificial muscles are often said to be analogous to biological muscles because they have force length curves and can produce force only in tension. This analogy is worth closer inspection. For example, the describes the passive force–length curve has been described as an

exponential function arising from connective tissue elements, while the active force–length relationship is typically modeled as a bell-shaped curve reflecting actin–myosin overlap [30]. These formulations emphasize that muscle force generation is governed by nonlinear geometric and material interactions rather than simple linear elasticity. McKibben type pneumatic artificial muscles (of which Festo is an example) exhibit analogous nonlinear behavior, though arising from fundamentally different physical mechanisms. As shown by Chou and Hannaford, the force–length relationship of McKibben actuators is derived from geometric constraints imposed by the braided shell, leading to a nonlinear dependence of force on actuator length through the braid angle [23]. In their formulation, actuator force is proportional to pressure and a nonlinear geometric term ($3 \cos^2 \theta - 1$), which governs both the maximum contraction limit and the curvature of the force–length relationship. The work presented here demonstrates that this nonlinear curve can be well approximated by an exponential as well.

However, there are some distinct differences between how these artificial muscles behave when compared to biological muscles. For example, the biological muscle optimum fiber length l_{OFL} is not equivalent to l_{rest} in the muscles. The maximum active force that the artificial muscles can produce is at the resting length, also the maximum length, of the artificial muscles. In contrast, the maximum active force a biological muscle length can produce is somewhere in the middle of the total contractile length of the muscle. Furthermore, biological muscles can adapt the optimal operating length through structural remodeling (e.g., sarcomere addition or tendon compliance) while BPAs have a fixed geometric configuration. This allows biological systems to self-optimize for the particular tasks the animal is subjected to. In order to re-optimize a robot for new tasks, new actuators would have to be swapped into the system. This distinction highlights a key limitation of BPAs in biomimetic applications, while also reinforcing the importance of accurate nonlinear modeling for control and design.

References

30. Winters, J.M. Hill-Based Muscle Models: A Systems Engineering Perspective. In Multiple Muscle Systems: Biomechanics and Movement Organization; Winters, Jack M.; Woo, Savio L.-Y., Eds.; Springer: New York, NY, 1990; pp. 63–93. https://doi.org/10.1007/978-1-4613-9030-5_5.

23. Chou, C.P.; Hannaford, B. Measurement and modeling of McKibben pneumatic artificial muscles. *IEEE Transactions on Robotics and Automation* 1996, 12, 90–102.
<https://doi.org/10.1109/70.481753>.

3. The figures, especially Figure 4, are quite small, the vertical scale reduced. There are few figures in the paper so I suggest they can afford to take up a little more space.

We have made Figures 2 through 4 larger for ease-of-readability.

4. An observation is that the disadvantage of BPAs are: "maximum tensile force occurs at resting (i.e., uninflated) length (l_{rest}) rather than at an optimal fiber length". Is the ability of the natural musculoskeletal system to adapt to the required loads and ranges placed upon them, by changing tendon length, sarcomere and fibre numbers, at least as significant problem when trying to mimic the natural system?

This is a great observation as well. We have included the following in Biomimetic Artificial Muscle Analogy subsection :

However, there are some distinct differences between how these artificial muscles behave when compared to biological muscles. For example, the biological muscle optimum fiber length l_{OFL} is not equivalent to l_{rest} in the muscles. The maximum active force that the artificial muscles can produce is at the resting length, also the maximum length, of the artificial muscles. In contrast, the maximum active force a biological muscle length can produce is somewhere in the middle of the total contractile length of the muscle. Furthermore, biological muscles can adapt the optimal operating length through structural remodeling (e.g., sarcomere addition or tendon compliance) while BPAs have a fixed geometric configuration. This allows biological systems to self-optimize for the particular tasks the animal is subjected to. In order to re-optimize a robot for new tasks, new actuators would have to be swapped into the system. This distinction highlights a key limitation of BPAs in biomimetic applications, while also reinforcing the importance of accurate nonlinear modeling for control and design.